

PS #03

Adaptive AI Tutor with Misconception Memory

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Adaptive AI Tutor with Misconception Memory (1/4)

Objective

- ❑ Teach a student across multiple sessions, not just answer questions
- ❑ Detect misconceptions and missing prerequisites from responses
- ❑ Generate adaptive hints, examples, quizzes, and revision plans
- ❑ Escalate to a teacher when confusion, distress, or cheating risk appears

Techniques

- ❑ Student model with concept mastery and confidence tracking
- ❑ Misconception graph linking weak topics to prerequisites
- ❑ Hint ladder: small clue → stronger hint → worked example
- ❑ Spaced repetition and adaptive difficulty scheduling
- ❑ Calibrated human handoff based on risk and repeated failure

Tools & frameworks

- ❑ LangGraph + OpenAI Agents SDK for tutor workflow
- ❑ LlamaIndex / LangChain for curriculum RAG
- ❑ Postgres + pgvector / Qdrant for learning memory
- ❑ Streamlit / Next.js for student and teacher views
- ❑ DeepEval / Ragas + Phoenix for learning and agent evaluation

Participants may choose alternative frameworks, libraries & models as per their preference.

Adaptive AI Tutor with Misconception Memory(2/4)

01 Requirements

- ☐ Students often get direct answers but not real understanding.
- ☐ Teachers cannot track every learner's hidden misconceptions in real time.
- ☐ One wrong answer may be treated the same as a deep concept gap.
- ☐ Generic practice ignores each student's pace, confidence, and weak prerequisites.

02 Agentic Challenge

- ☐ The tutor must diagnose whether an error is careless, conceptual, or prerequisite-based.
- ☐ It must adjust explanation style, difficulty, hint level, and practice automatically.
- ☐ It should update long-term learning memory without overreacting to temporary mistakes.
- ☐ It must escalate when repeated confusion, distress, cheating intent, or low confidence appears.

03 Final Output

- ☐ A personalized tutoring session with diagnostic questions and adaptive hints.
- ☐ A misconception profile showing weak concepts and missing prerequisites.
- ☐ Teacher dashboard with progress, risk, and escalation reasons.
- ☐ Evidence of learning gain from pre-test to post-test across multiple rounds.

Adaptive AI Tutor with Misconception Memory (3/4)

Technical Architecture & Evaluation

Student + Teacher UI

Streamlit / Next.js dashboards for session, progress, and escalation

Tutor Orchestrator

LangGraph state machine + OpenAI Agents SDK for tutor roles

Curriculum Knowledge

LlamaIndex / LangChain RAG over lessons, rubrics, and examples

Learning Memory

Postgres + pgvector/Qdrant for mastery, misconceptions, and evidence

Guardrails + Evals

Hint policy, deferral rules, Phoenix tracing, DeepEval/Ragas

Judging Metrics

Use metrics to show real tutoring improvement, not just polished answers.

Learning gain

pre-test to post-test improvement

Misconception recall

finds real weak concepts

Adaptation quality

difficulty and hints adjust

Escalation precision

handoff only when needed

Memory usefulness

past sessions improve help

Hint quality

guides without answer dump

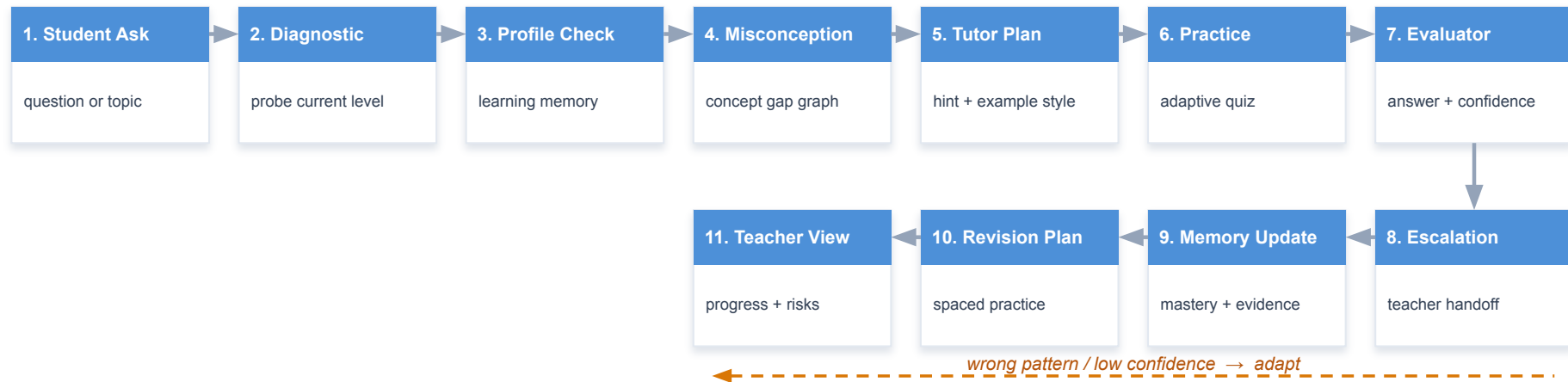
Demo must show one learning loop

Example: student misunderstands fractions → tutor diagnoses prerequisite gap → adapts practice → updates memory or escalates.

Adaptive AI Tutor with Misconception Memory (4/4)



Student asks: "I don't understand fractions. Why is $1/2$ bigger than $1/3$?"



Misconception
Memory

Adaptive Teaching

Hints Before Answers

Calibrated Confidence

Teacher Escalation

Tutoring policy: guide with hints before answers • update memory from repeated evidence • escalate for persistent confusion, distress, cheating, or low confidence